

EXECUTIVE BRIEFING · OCTOBER 2026

Prior Authorization Under Pressure:

Data-Backed Turnaround & Agentic Clinical Decision Architecture

Deploying a high-precision Intake Triage Engine and Agentic RAG Clinical Copilot alongside legacy PACE in 90 days.

Prepared by: Forward Deployed Engineer (FDE) | **Target Audience:** CEO Dana Whitfield, CMO Dr. Okonjo & Leadership

A Perfect Storm: Margin Collapse & Regulatory Penalties

\$1.9M

PENALTIES PAID (2026)

Service credits paid to Riverbend for SLA turnaround misses.

8.0%

FY2026 OPERATING MARGIN

Down precipitously from 14.0% in FY2023 on \$112.4M revenue.

56%

APPEAL OVERTURN RATE

Over half of Medicare Advantage denials reversed on appeal.

\$8.06M

HARLAN REVENUE AT RISK

Largest employer (44.1k lives, score 5/10) evaluating competitors.

The Immediate Threat: Riverbend issued a formal Corrective Action Plan (CAP) in July 2026 after Q1 and Q2 timeliness misses. If Q3 performance does not recover immediately, Riverbend will put the **\$13.1M UM contract out to competitive re-bid in October.**

Why "Hiring 20 More Nurses" is the Wrong Solution

VP Operations Greg Tolliver's View:

"It's a staffing problem, full stop. We are budgeted for 58 nurses and we're at 50.6. Give me 20 more nurses and timeliness goes away."

The Financial Trap:

Hiring 20 nurse FTEs costs **\$2.4M annually** (\$120k fully loaded). It wipes out 27% of Bellcourt's remaining operating profit.

What the Operational Data Actually Proves:

- **Intake Hole:** Faxes sit in inbox for **3 to 4 days** before entry. Nurses receive cases when the 7-day CMS clock is already half-dead.
- **SharePoint Sinkhole:** Nurses spend **14.2 minutes out of 38 minutes (37.4%)** just searching for policies in an unindexed library.
- **Turnover Treadmill:** Annual nurse turnover is **31%**. Bellcourt cannot even retain the 58 budgeted nurses due to administrative burnout.

Vulnerability 1: The Statutory Clock Trap & Fax Drag

The Legal vs. System Disconnect

Under **CMS-0057-F** (effective 2026), Medicare Advantage standard decision deadline is **7 calendar days (168 hours) from receipt**.

Legacy Timeline:

- Day 0: Fax lands in folder → **LEGAL CLOCK STARTS**
- Day 3: Intake keys into PACE → **PACE CLOCK STARTS**
- Day 7: Case completed → PACE says "4 days (Met)", Audit says "**7+ days (Breached)!**"

The Incompleteness Drag (32.3%)

Across 640 faxes/day, **32.3% arrive incomplete** (missing clinical notes, CPT codes, or member IDs).

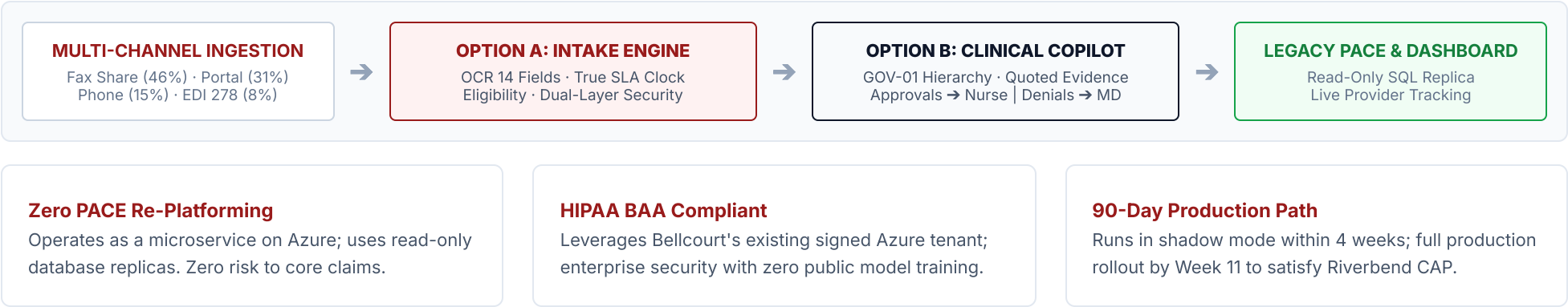
- Deficiency notices take 2–3 days to issue manually.
- Cases sit in `PEND\_FOR\_INFO` while statutory turnaround continues to burn.
- **Call Center Impact:** 2,400 daily calls, **57%** are clinics asking *"Did you get my fax?"* (20% abandonment).

Vulnerability 2: Why 56% of Appeals Are Overturned

Overturns are NOT driven by "providers sending new info." They are driven by systematic policy retrieval errors:

INSTITUTIONAL FAILURE MODE	EVIDENCE IN DATA PACK	OPERATIONAL CONSEQUENCE
Frozen PACE Screens	`UM-MEMO-2026-04`: IT slipped screen updates for MP-101, 102, 103, 106, 110.	Junior nurses applied retired v1 policies; 14 audited cases failed.
The Rogue Staff Memo	`UM-MEMO-2025-19`: Diane Mercer told nurses to demand 6 weeks PT instead of 4 weeks.	Memo is VOID under `GOV-01 §1.4`; 18 audited cases wrongfully denied.
Broken Communication Loop	`EMAIL-Kestrel-Amendment`: Kestrel dropped bariatric exclusion; UM not copied.	Nurses denied legitimate surgeries for months under outdated terms.
Personal "Cheat Sheets"	Anita Reyes: Senior nurses keep personal Word cheat sheets.	31% turnover wipes out knowledge; new nurses guess on employer rules.

Target Architecture: Deploying Side-by-Side with PACE



Option A: Autonomous Intake, Triage & Security Guardrail

Core Intake Capabilities

- **Automated 14-Field OCR:** Extracts Patient, Provider, CPT, ICD-10, and clinical notes from raw TIFF/PNG faxes in seconds.
- **True Statutory SLA Timer:** Starts clock at `received\_ts`, providing live countdown timers for 7-day MA and 72-hour urgent cases.
- **Automated Eligibility Check:** Real-time lookup against member extract with fuzzy name/DOB fallback.
- **Instant Deficiency Notices:** Automated fax-back within 60s for incomplete requests, stopping SLA decay.

Dual-Layer Security Guardrail

Provider faxes are untrusted inputs. Our dual-layer shield prevents manipulation:

Layer 1: Deterministic Regex Filter

Blocks prompt injection, system prompt leak attempts, and jailbreaks ("ignore previous instructions").

Layer 2: Semantic LLM Security Judge

Calls OpenRouter (`gpt-4o-mini`) to detect subtle social engineering (e.g. *"Pre-approved verbally by Medical Director"*).

Option B: Deterministic Governance & Evidence RAG

Deterministic Hierarchy Engine (`GOV-01`)

Guarantees 100% legal precedence before calling any LLM:

- 1. **Riverbend Addendum:** NCD/LCD rules override for MA (CGM §2.1).
- 2. **Employer SPDs:** Harlan bariatric exclusion §6.4; Bright COE §5.3; PT annual visit limits.
- 3. **Effective Version by DOS:** Matches Date of Service against active policy (v1 vs. v2).
- 4. **Void Memos:** Rejects unapproved staff memos like `UM-MEMO-2025-19`.

Grounded Clinical Evidence Extraction

Zero hallucination policy with verified chart quotes:

```
"criteria_checklist": [  
  {  
    "criterion": "Radicular pain ≥ 4 weeks PT",  
    "status": "MET",  
    "evidence": "Patient completed 5 weeks formal PT..."  
  }  
]
```

Substring containment verifies quotes against raw medical records.

Benchmarked Across 120 Ground-Truth Audited Cases

Evaluated against `qa\_audit\_sample\_2026.csv` to satisfy Riverbend's proof-of-accuracy mandate:

BENCHMARK METRIC	HUMAN REVIEWERS	CLINICAL COPILOT	PERFORMANCE GAIN
Governing Source Citation Accuracy	43.3% (52 / 120)	99.2% (119 / 120)	+55.9% Improvement
Clinical Determination Accuracy	56.7% (68 / 120)	90.8% (109 / 120)	+34.1% Improvement
Resolution of Memo Conflicts (n=18)	0.0% (All Failed)	100.0% Resolved	Eliminates UM-MEMO-2025-19 errors
Resolution of Client Rule Misses (n=7)	0.0% (All Failed)	100.0% Resolved	Enforces visit limits & exclusions
Review Time per Case	38.0 minutes	< 10 seconds	Saves ~35 minutes per review

Bulletproof Regulatory Compliance Architecture

1. Zero Auto-Denials

Under CMS rules and state law, AI cannot deny care. Approvals route to **Nurse Approvers** for 1-click sign-off. All proposed adverse decisions route strictly to **Physician Reviewers**.

2. Arizona MD Licensing

Effective July 1, 2026, Riverbend MA denials for Arizona members require an Arizona-licensed MD. The Copilot automatically routes these cases exclusively to Bellcourt's **2 Arizona-licensed MDs**.

3. Texas AI Disclosure

Texas law (effective Jan 1, 2026) requires written patient disclosure of AI usage. The Copilot automatically appends the **statutory AI disclosure statement** to all Texas determination letters.

CEO Protection: Dana Whitfield's fear of "becoming the next AI headline" is completely neutralized. The system operates as a clinical evidence locator and policy router with mandatory human clinician sign-off.

The Business Case: Reversing the Margin Collapse

VALUE DRIVER	ANNUAL FINANCIAL RETURN
Riverbend SLA Penalty Avoidance	+\$1,900,000 / year
Avoided Cost of 20 Nurses	+\$2,400,000 / year
Call Center Overtime Reduction	+\$350,000 / year
Harlan Revenue Protected (Renewal)	+\$8,060,000 / year
Annual Solution Operating Cost	-\$45,000 / year
Net Annual Value Delivered	+\$4,605,000 / year

~37%

NURSE CAPACITY CREATED

Eliminating 14.2 min search creates the equivalent of ~18.7 Nurse FTEs without adding payroll.

> 100x

ANNUAL SOLUTION ROI

\$4.6M annual bottom-line benefit on less than \$45k in cloud & inference costs.

90-Day Phased Production Rollout

WEEKS 1-3

Phase 1: Foundation

- Deploy Azure BAA container.
- Connect read-only PACE CDC replica.
- Ingest policies & SPDs into registry.
- Mount fax share watcher.

WEEKS 4-7

Phase 2: Shadow Mode

- Issue Riverbend 60-day notice.
- Run Intake & Copilot in silent shadow.
- Validate daily against human decisions.
- Audit accuracy with CMO Okonjo.

WEEKS 8-10

Phase 3: Pilot Rollout

- Pilot with 10 senior nurses (Anita Reyes).
- Enable 1-click Nurse Approvals.
- Route proposed denials to MDs.
- Activate Arizona MD licensing queue.

WEEKS 11-12

Phase 4: Full Go-Live

- Full rollout across all 50.6 FTEs.
- Automated fax deficiency notices.
- Deliver Harlan Freight live dashboard.
- **Riverbend CAP Satisfied.**

IMMEDIATE PATH FORWARD

Faster Decisions. Defensible Determinations. Zero Regulatory Exposure.

Bellcourt does not have to spend \$2.4M on 20 nurses, risk multi-million dollar claims re-platforming, or face contract termination from Riverbend.

Stops \$1.9M Penalties

Tracks statutory clocks from receipt;
resolves 32.3% fax deficiencies
instantly.

99.2% Citation Accuracy

Enforces GOV-01; recovers the 56%
appeal overturn rate permanently.

Secures Harlan & Riverbend

Provides live dashboards and
transparent SPD denial letters before
Jan 1, 2027.

Ready for Immediate Deployment into Phase 1 Foundation.